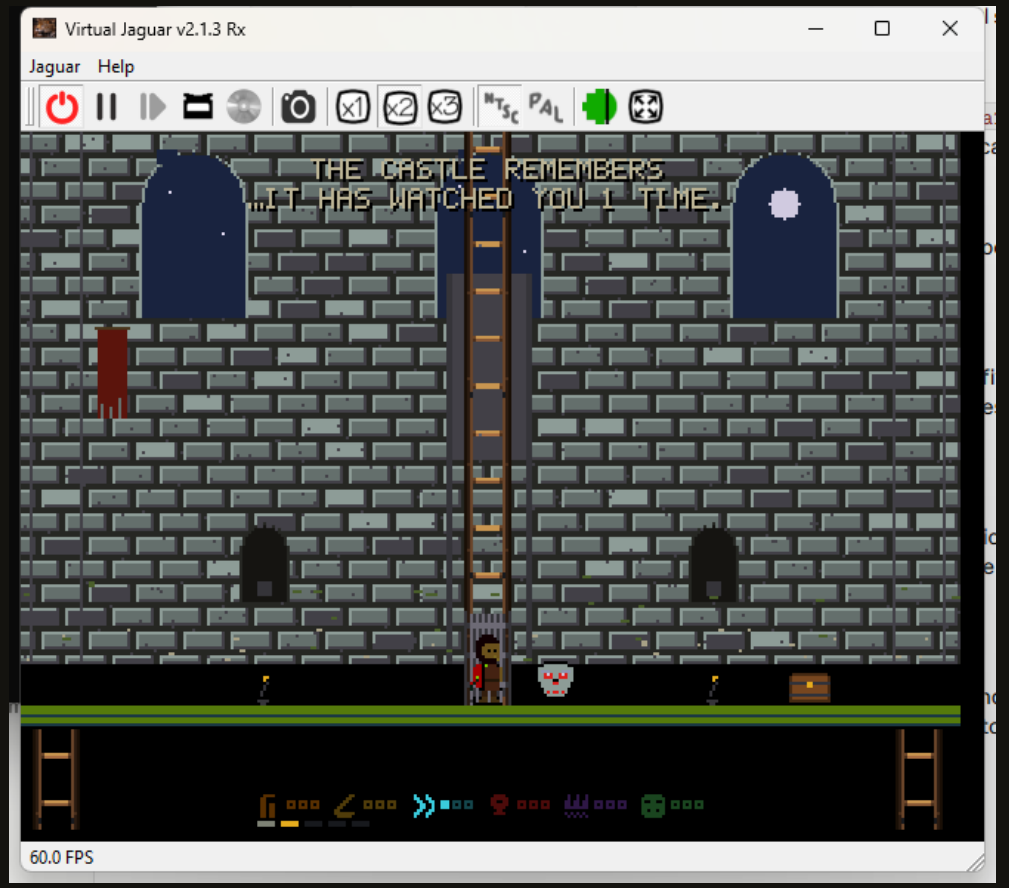
Castle Remembers Jaguar

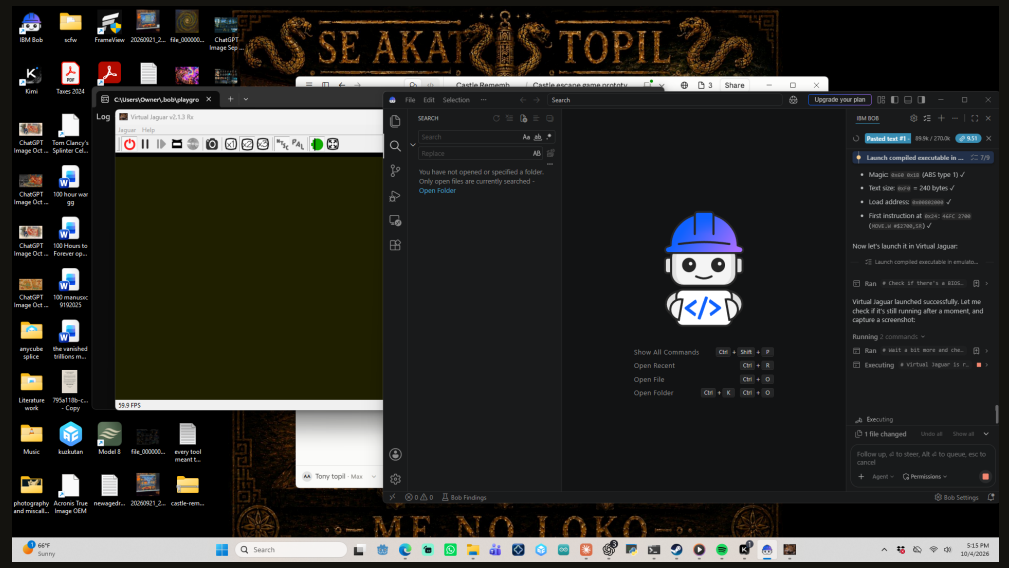
- Tony Topil · creator and project lead
- A native Atari Jaguar port, in 68000 assembly, of a game whose castle remembers how you play.
- Track: Theme 2 · Modernize What Matters



Virtual Jaguar capture · Gate 7

Keep the idea. Rebuild the machine.

- The original is a single-file HTML5 game: the castle watches the player and adapts rooms, traps and enemies.
- The port replaces browser rendering, input and audio with Jaguar object lists, CRY16 sprites and controller registers, all in Motorola 68000 assembly.
- The HTML and Python/Pygame editions are separate projects. Jaguar is this entry.

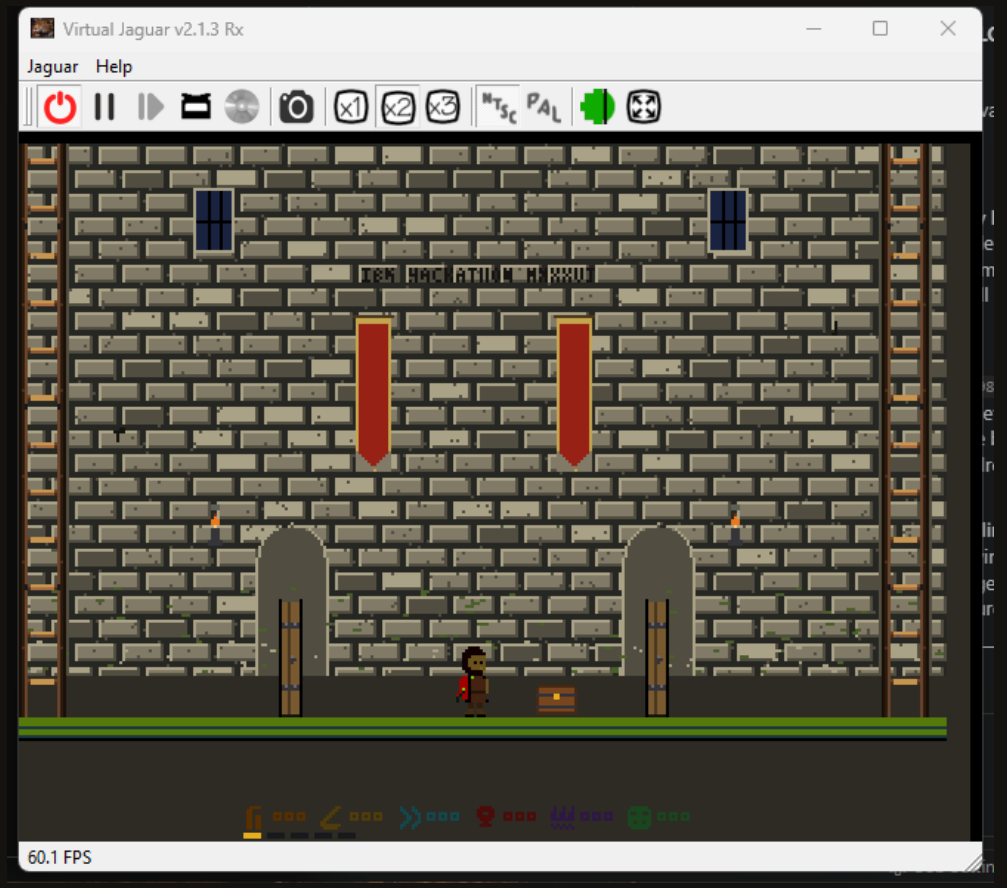


Gate 1: first native frame, Virtual Jaguar

PROJECT SUMMARY

A five-floor castle that adapts

- Five floors with doors, levers, gates, chests, a long ladder, six enemy types, spikes, falling masonry and a blade trap.
- A six-category memory records where you hide, how you die and what you open. On each death the castle rebuilds itself and adapts enemy spawns and speed.
- A castle-voice text band and icon HUD make the memory visible. The rules are deterministic: same habits, same response.



Gate 7, floor 1 · Virtual Jaguar

What IBM Bob contributed

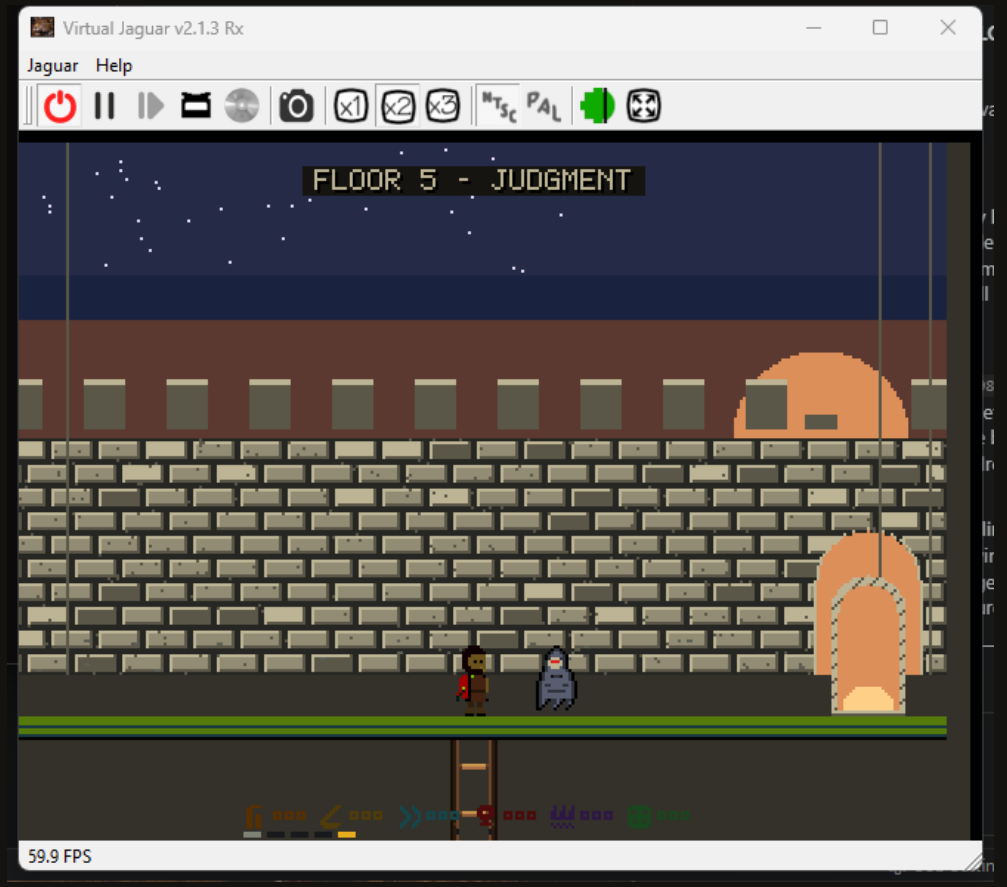
- Native Windows build path: RMAC 2.5.2, RLN 1.7.7, Virtual Jaguar Rx R5.
- Rendering, input and adaptive-memory foundation (Gates 1–5, each recorded as a runtime pass).
- Architecture reviews of the text object and buffer (Checkpoint B) and of resident backgrounds and object order (Checkpoint C).
- Claude handled later gameplay; Kimi produced art. All attribution is in docs/submission/EVIDENCE.md.



Hero ladder-climb frames · Virtual Jaguar

From blue screen to five floors

- Oct 4: Windows toolchain, first blue screen, preserved HTML baseline.
- Oct 5: hero sprite, controller movement, floor and jump, collision, adaptive enemy; then the Gate 7 five-floor castle with interactions, traps and expanded memory.
- Oct 5–7: castle voice and HUD, ladders, five resident environment bands, hero walk and climb frames.
- Oct 18: submitted build merged to main; browser demo page added.



Gate 7, floor 5 · Virtual Jaguar

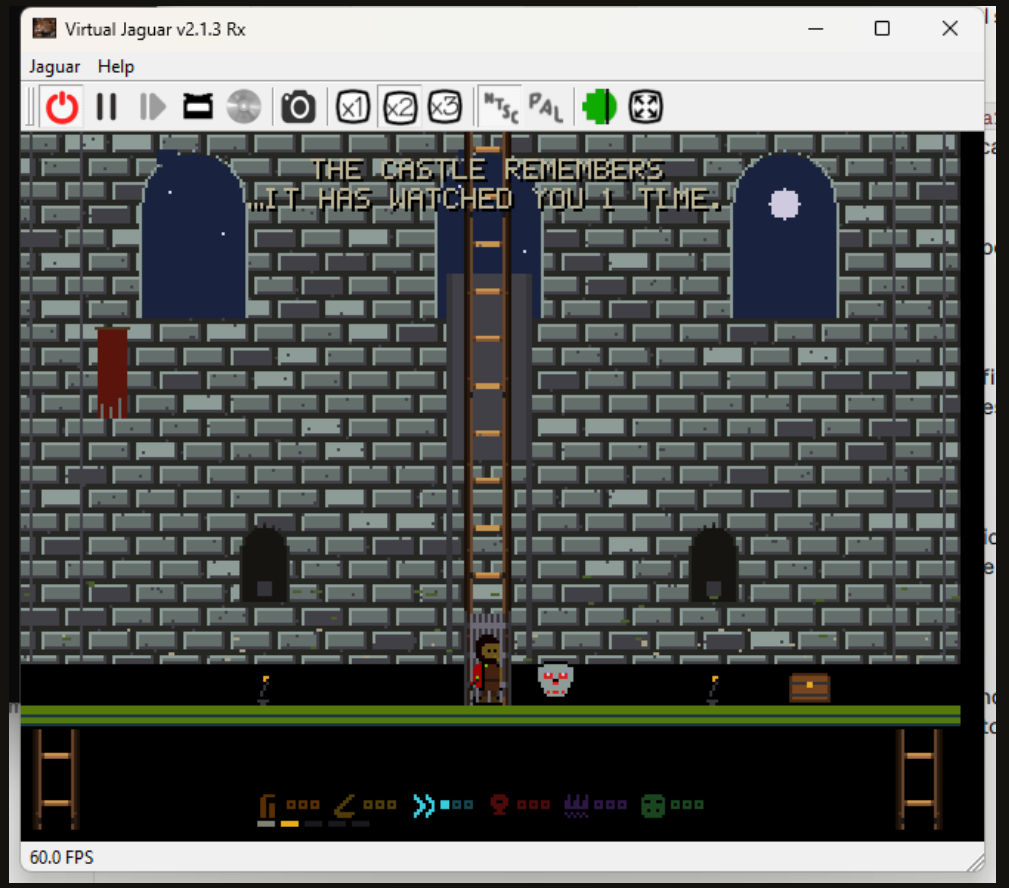
Evidence, and its limits

- Gates 1–5: recorded runtime passes in Virtual Jaguar. Gate 7 normal build: 59.9–60 FPS, COF header guard passing.
- Scripted playtests run the real 68000 code in a Python harness; Oct 13 checks passed a full five-floor route plus boot, controls, bands and climb scenarios.
- The harness is a partial hardware model, not full emulation. No physical Jaguar testing and no runtime AI are claimed.
- Gate 6 runtime verification and Bob's broader low-level audit remain open.

TRY IT

Live demo, source and video

- Live demo (free, no sign-in): theworldasiknowit777.github.io/castle-remembers-jaguar/play/
- Runs the real gate7_castle.cof in a browser Jaguar emulator (EmulatorJS, Virtual Jaguar core).
- Code, README, build and test steps: github.com/theworldasiknowit777/castle-remembers-jaguar
- Demo video (3 min or less): YouTube link in the submission form.



Gate 7 · Virtual Jaguar

Next steps

- Validate on physical Jaguar hardware and close Bob's remaining low-level audit items.
- Verify Gate 6 at runtime, and pin tool and dependency versions for fully reproducible builds.
- Build an automated browser-to-Jaguar behaviour-equivalence harness.
- Learnings: explicit object and memory reviews make a legacy-console port testable; preserving the source makes attribution defensible.